Math 9/10 - Reasoning and Analyzing Units (5 + 6) v2

Rubric								
Outcome	0	1	2	3	4	5	6	7
Reasoning and Analyzing								

Name : _____

Teacher's name: _____

1. For each of the following statements, identify if the statement is **always true**, **sometimes true**, or **never true**. Provide a brief justification.

   (a) If $ax^n + bx^m$ is a polynomial, then the leading coefficient is a.

   (b) If θ is an acute angle in a right triangle, then $\cos\theta > 1$.

   (c) For non-zero values a, b, c, d, if $\dfrac{a+b}{c} = \dfrac{d}{a-b}$, then $a^2 - b^2 = cd$.

   (d) Multiplying a binomial by a binomial results in a trinomial.

   (e) If you double the lengths of all sides of a right triangle, the sine of its acute angles will also double.

2. A student was asked to expand and simplify the expression $(2x-1)^2-(x+4)$. Which step did they **first** make an error? Explain the error.

$$\begin{aligned}(2x-1)^2-(x+4) &= (4x^2-4x+1)-(x+4) && \textbf{Step 1}\\ &= 4x^2-4x+1-x+4 && \textbf{Step 2}\\ &= 4x^2-5x+5 && \textbf{Step 3}\end{aligned}$$

3. In each of the following, one item doesn't belong.

   (a) Which trigonometric ratio doesn't belong? Explain your reasoning.

   i. $\sin(60°)$ ii. $\tan(30°)$ iii. $\tan(60°)$ iv. $\cos(45°)$

   (b) Which polynomial doesn't belong? Explain.

   i. x^2-25 ii. $x^2+8x+16$ iii. $2x^2+10x$ iv. x^2+4x+5